Physics 4700H: Assignment VII-A

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Problem 1. Consider a system consisting of two particles.

(a) First, assume the particles are distinguishable, and label them A and B. Suppose each particle can be in one of three states, where for A these have energies α_1, α_2, and α_3 whereas for B they have energies β_1, β_2, and β_3. Assume that the state of A is independent of the state of B and vice versa. Show that the partition function for the combined system $Z_{tot} = Z_A Z_B$.

(b) Now, assume the particles are indistinguishable. Again, each particle can be in one of three states, with energies ϵ_1, ϵ_2, and ϵ_3. The particles are not interacting, and there is no restriction on the number of particles in a given state. What is the partition function of the combined system in this case?

Solution 1.

(a) First note that by our usual definition the individual partition functions are

$$Z_A = e^{-\alpha_1/k_BT} + e^{-\alpha_2/k_BT} + e^{-\alpha_3/k_BT}$$

and

$$Z_B = e^{-\beta_1/k_BT} + e^{-\beta_2/k_BT} + e^{-\beta_3/k_BT}.$$

The combined system of these particles is allowed to have energies corresponding to any sum of the individual particle energies. Defining the partition function as usual then this looks like

$$Z_{tot} = \sum_i \sum_j e^{-(\beta_i+\alpha_j)/k_BT}.$$

We can write out all of these terms, there are $3 \cdot \binom{3}{1} = 9$ terms, three groups where we choose one β and then select all αs,

$$\begin{aligned} Z_{tot} = {} & e^{-(\beta_1+\alpha_1)/k_BT} + e^{-(\beta_1+\alpha_2)/k_BT} + e^{-(\beta_1+\alpha_3)/k_BT} \\ & + e^{-(\beta_2+\alpha_1)/k_BT} + e^{-(\beta_2+\alpha_2)/k_BT} + e^{-(\beta_2+\alpha_3)/k_BT} \\ & + e^{-(\beta_3+\alpha_1)/k_BT} + e^{-(\beta_3+\alpha_2)/k_BT} + e^{-(\beta_3+\alpha_3)/k_BT}. \end{aligned}$$

Now we look at $Z_A Z_B$ (I switched them at the first equals sign to make it look more like the above result),

$$\begin{aligned} Z_A Z_B &= (e^{-\beta_1/k_BT} + e^{-\beta_2/k_BT} + e^{-\beta_3/k_BT})(e^{-\alpha_1/k_BT} + e^{-\alpha_2/k_BT} + e^{-\alpha_3/k_BT}) \\ &= e^{-\beta_1/k_BT}(e^{-\alpha_1/k_BT} + e^{-\alpha_2/k_BT} + e^{-\alpha_3/k_BT}) \\ &\quad + e^{-\beta_2/k_BT}(e^{-\alpha_1/k_BT} + e^{-\alpha_2/k_BT} + e^{-\alpha_3/k_BT}) \\ &\quad + e^{-\beta_3/k_BT}(e^{-\alpha_1/k_BT} + e^{-\alpha_2/k_BT} + e^{-\alpha_3/k_BT}) \\ &= e^{-(\beta_1+\alpha_1)/k_BT} + e^{-(\beta_1+\alpha_2)/k_BT} + e^{-(\beta_1+\alpha_3)/k_BT} \\ &\quad + e^{-(\beta_2+\alpha_1)/k_BT} + e^{-(\beta_2+\alpha_2)/k_BT} + e^{-(\beta_2+\alpha_3)/k_BT} \\ &\quad + e^{-(\beta_3+\alpha_1)/k_BT} + e^{-(\beta_3+\alpha_2)/k_BT} + e^{-(\beta_3+\alpha_3)/k_BT}, \end{aligned}$$

indeed, the same answer, so $Z_{tot} = Z_A Z_B$.

(b) In this case the partition functions for the two particles are identical,

$$Z_1 = e^{-\epsilon_1/k_BT} + e^{-\epsilon_2/k_BT} + e^{-\epsilon_3/k_BT}.$$

The entire system then, as before, can have energies corresponding to any sum of ϵs and so has partition function

$$Z_{tot} = \sum_i \sum_j e^{-(\epsilon_i+\epsilon_j)/k_BT}.$$

Like before, this will have nine terms. However, since ϵ_i could be the same as ϵ_j, not all of the terms are unique. Writing them all out we see that

$$\begin{aligned} Z_{tot} &= e^{-(\epsilon_1+\epsilon_1)/k_BT} + e^{-(\epsilon_1+\epsilon_2)/k_BT} + e^{-(\epsilon_1+\epsilon_3)/k_BT} \\ &\quad + e^{-(\epsilon_2+\epsilon_1)/k_BT} + e^{-(\epsilon_2+\epsilon_2)/k_BT} + e^{-(\epsilon_2+\epsilon_3)/k_BT} \\ &\quad + e^{-(\epsilon_3+\epsilon_1)/k_BT} + e^{-(\epsilon_3+\epsilon_2)/k_BT} + e^{-(\epsilon_3+\epsilon_3)/k_BT} \\ &= e^{-2\epsilon_1/k_BT} + e^{-2\epsilon_2/k_BT} + e^{-2\epsilon_3/k_BT} \\ &\quad + 2e^{-(\epsilon_1+\epsilon_2)/k_BT} + 2e^{-(\epsilon_1+\epsilon_3)/k_BT} + 2e^{-(\epsilon_2+\epsilon_3)/k_BT} \end{aligned}$$

Problem 2. Consider a classical (i.e. high temperature and/or low density) gas of N indistinguishable non-interacting spin-1 particles, each of mass m, confined to a two-dimensional box of size L^2. What is the entropy at temperature T? Note that spin-1 particles have three possible spin states, which can be labeled $\uparrow$, 0, and $\downarrow$. There is no applied magnetic field in this problem, so these spin states all have the same energy. You may define that spin energy to be zero.

Solution 2.